

ARCHANA VERMA

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OBJECTIVE

Ph.D. Chemical Engineer with experience leading cross-functional research projects, process scale-up, and industrial collaborations. Proven ability to manage multidisciplinary programs, align stakeholders, and translate complex concepts for diverse audiences. Combines technical expertise with record of leadership, outreach, and relationship-building to translate scientific insight into strategic impact.

EDUCATION

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Ph.D. | Chemical Engineering | May 2026

GPA: 3.95

M.S. | Chemical Engineering | May 2024

Fellowships: NSF Graduate Research Fellow, Illinois Distinguished Fellow, Samuel W. Parr Fellow

STANFORD UNIVERSITY

B.S. | Chemical Engineering | June 2021

GPA: 3.86

RESEARCH EXPERIENCE

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Graduate Research Assistant | Prof. Nicholas E. Jackson | Theoretical Materials Chemistry | Sept. 2021 - May 2026

- Created a computational platform from scratch integrating quantum chemistry and film morphology to predict doping behavior in organic semiconductors where existing models could not capture multiscale phenomena, translating simulation results into design guidelines for improving semiconductor doping efficiency
- Collaborated with experimental team to expedite polymer design with machine learning, identifying relevant molecular descriptors and designing analysis workflows to translate data into actionable materials design recommendations; initiated a cross-functional subgroup on doping to foster collaboration within the lab
- Simultaneously mentored an undergraduate student for four years in an independent research project focused on challenging conventional assumptions of how disorder affects charge transfer using quantum mechanical analysis (DFT)

STANFORD UNIVERSITY

Undergraduate Research Assistant | Prof. Gerald G. Fuller | Complex Fluids | Sept. 2019 - June 2021

- Consulted with Johnson & Johnson Vision Care to evaluate new product performance by correlating contact angle with tear film stability using MATLAB image analysis, presenting data-driven recommendations to diverse stakeholders

Undergraduate Research Assistant | Prof. Michael D. Fayer | Physical Chemistry | Sept. 2017 - June 2019

- Led the development and characterization of novel material frameworks to investigate fundamental molecular mechanisms of proton transport, applying state-of-the-art 2D spectroscopy to solve complex chemical challenges

INTERSHIPS

BASF

PhD Leadership Development Program Intern | Process Operations Analyst | June 2025 - Aug. 2025

- Re-designed lubricant formulation to enhance conductivity to meet client specifications, identifying additive candidates across five chemical classes designed to isolate structure-property relationships
- Scaled selected formulations from laboratory batches to pilot-scale trials, using monthly performance monitoring data to validate formulation stability and process feasibility in coordination with international manufacturing teams

CORTEVA AGRISCIENCE

R&D Intern | Formulations Chemistry | June 2020 - Aug. 2020

- Bridged chemistry and informatics teams to automate product development workflow, designing an accessible software tool for high-throughput solvent screening that enabled chemists to streamline experimental validation

REGENERON PHARMACEUTICALS

R&D Intern | Analytical Chemistry | June 2017 - Aug. 2017

- Optimized protocol parameters to improve contaminant detection in pre-clinical drug testing of monoclonal antibodies

OUTREACH & LEADERSHIP

Graduate Society of Women Engineers | weSTEM Director, Speaker Chair, Social Media Chair | 2023 - 2026

- Headed executive management of 11-member team to organize national conference (130 attendees & 5 corporate partners)
- Remained \$6,000 under budget target while expanding conference programming events, doubling external sponsor investment, awarding a record number of scholarships, and recruiting the first international attendees
- Directed end-to-end recruitment of 23 speakers overseeing selection, contracts, travel, scheduling, and risk mitigation
- Grew online reach (3,400%) and engagement (2,200%) by creating and leading video team in a digital branding strategy
- Earned Grassroots Initiatives to Address Needs Together Leadership Award for proposal evaluating community impact

Crisis Text Line | Volunteer Crisis Counselor | 2023 - Present

- De-escalated 250+ individuals in high-pressure situations using adaptive communication and collaborative problem-solving
- Completed 50+ hours of training in evidence-based conflict resolution, goal alignment, and action planning frameworks

AAAS Catalyzing Advocacy in Science and Engineering (CASE) Workshop | Attendee | 2026

- Selected as the Graduate College representative and one of four UIUC students to advocate on Capitol Hill for federal science funding, communicating the impact of research investment to policymakers

PRESENTATIONS & AWARDS

University Archives Women in Science Lecture | Oral (Invited) | 2026
American Institute for Chemical Engineers (AIChE) Annual Conference | Poster | 2020, 2025
- 2025: Women in Chemical Engineering Community Travel Award winner
Gender Equity and Inclusion in Chemical Sciences Conference | Panelist | 2024
- Gender Equity and Inclusion in Chemical Sciences Inclusive Leadership Award Winner
American Physical Society (APS) March Meeting | Oral | 2024
Midwest Thermodynamics and Statistical Mechanics Conference | Poster | 2023 | Oral | 2024
UIUC Chemical Engineering Graduate Student Symposium | Poster | 2023 | Oral | 2024
- 2024: First Place Oral Presentation
American Chemical Society (ACS) Central Regional Meeting | Oral (Invited) | 2023
Regeneron Science Talent Search | Oral | 2017
- Fifth Place Overall Nationally
University of Toronto International Research Invitational | Oral (Invited) | 2017
Intel International Science and Engineering Fair | Poster | 2016, 2017
- 2017: Second Place in Chemistry, ACS Third Place Special Award
- 2016: First Place in Chemistry
New York State Science and Engineering Fair | Poster | 2016, 2017
- 2017: First Place in Chemistry, Millennial Materials Special Award
- 2016: First Place in Chemistry
Long Island Science and Engineering Fair | Poster | 2016, 2017
- 2017: First Place in Chemistry & Materials Science, ASM Materials Science Special Award
- 2016: First Place in Chemistry, Yale Science and Engineering Association Special Award
National Junior Science and Humanities Symposium | Poster | 2016
- Second Place in Chemistry Poster Division

TEACHING EXPERIENCE

SELF-EMPLOYED

Private Tutor and Essay Consultant | Spring 2026 - Present

- Delivered strategic feedback for 100+ academic essays, leading to selective admissions and competitive award outcomes
- Designed individualized tutoring lesson plans for standardized test preparation, and quantitative STEM coursework

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Graduate TA | Data Science for Chemistry and Engineering | Fall 2023

- Independently led weekly office hours, graded coding assignments, and was responsible for course communications

Graduate TA (Technical Writing Team) | Process Design | Fall 2022

- Co-led weekly discussion sections and office hours focused on enhancing technical communication for capstone design

STANFORD UNIVERSITY

Undergraduate TA | Foundations of Physical Chemistry | Spring 2021

- Created course homeworks and exams, held weekly office hours, managed communications, grading, and Q&A forum

Undergraduate TA | Physical Chemistry I: Thermodynamics | Spring 2020

- Co-led two weekly computational labs, held weekly office hours, managed communications, grading, and Q&A forum

PUBLICATIONS

* denotes equal contribution

- [8] Verma, A., Wang, C.-I., et al. (2025). *Journal of Physical Chemistry A*. <https://doi.org/10.1021/acs.jpca.5c05040>
- [7] Raghuraman, V., Verma, A., & Jackson, N.E. (2025). *Journal of Chemical Theory and Computation*. <https://doi.org/10.1021/acs.jctc.5c01206>
- [6] Xu, Z.*, Wang, Z.*, Zhang, R.*, Verma, A.*, et al. (2025). *Journal of the American Chemical Society*. <https://doi.org/10.1021/jacs.5c07995>
- [5] Verma, A., & Jackson, N. E. (2024). *Journal of Chemical Physics*. <https://doi.org/10.1063/5.0197816>
- [4] Suja, V. C.*, Verma, A.*, et al. (2022). *Journal of Colloid and Interface Science*. <https://doi.org/10.1016/j.jcis.2022.01.075>
- [3] Verma, A., Stoppelman, J.P., & McDaniel, J.G. (2020). *International Journal of Molecular Sciences*. <https://doi.org/10.3390/ijms21020403>
- [2] McDaniel, J. G., & Verma, A. (2019). *Journal of Physical Chemistry B*. <https://doi.org/10.1021/acs.jpcc.9b02187>
- [1] Carbery, W. P., Verma, A., & Turner, D. B. (2017). *Journal of Physical Chemistry Letters*. <https://doi.org/10.1021/acs.jpclett.7b00130>

SKILLS & COURSEWORK

Project Management: Cross-Functional Collaboration, Stakeholder Management, Process Improvement, Product Development

Technical Communication: Public Speaking, Scientific Writing, Scientific Visualization, Curriculum Development, Tutoring

Programming & Software: Python, C/C++, MATLAB, Mathematica, Java, NWChem, LAMMPS, Avogadro

Scientific: Molecular Simulation, Data Science, Physical Chemistry, Polymer Physics, Optical Spectroscopy

Selected Coursework: Chemical Processes in Energy Systems, Bioenergy & Biofuels, Transportation Equity, Engineering Leadership